Illustration 9.5

Calculate $\lim\limits_{x\to\infty}\dfrac{2x^2-2x+3}{x^2+4x+4}$.

Solution

If we could try to use limit theorems to calculate this limit, we end up in the following situations.

$(2x^2-2x+3)\to\infty$ as $x\to\infty$

$(x^2+4x+3)\to\infty$ as $x\to\infty$

$\lim\limits_{x\to\infty}\left(\dfrac{2x^2-2x+3}{x^2+4x+3}\right)=\dfrac{\infty}{\infty}$ which is called an indeterminate form.

But actual calculation and tabulation gives the following :

x	$2x^2-2x+3$	x^2+4x+4	$\dfrac{2x^2-2x+3}{x^2+4x+4}$
1	3	9	0.3333
10	183	144	1.27083
100	19803	10404	1.90340
1000	1998003	1004004	1.99003
10000	199980003	100040004	1.99900

Table values show that as x becomes sufficiently large, $f(x)$ becomes closer and closer to 2. Then,

$\lim\limits_{x\to\infty}\left(\dfrac{2x^2-2x+3}{x^2+4x+4}\right)=2.$

Fortunately, we may simplify the problem by dividing the numerator and denominator by x^2. We have

$$\lim_{x\to\infty}\frac{2x^2-2x+3}{x^2+4x+4}=\lim_{x\to\infty}\left(\frac{2-\frac{2}{x}+\frac{3}{x^2}}{1+\frac{4}{x}+\frac{4}{x^2}}\right)$$

$$=\frac{2-0+0}{1+0+0}\quad\left(\text{since }\frac{1}{x}\to 0 \text{ as } x\to\infty,\ \frac{1}{x^2}\to 0 \text{ as } x\to\infty\right)$$

$$=2.$$

Note that the degree of both numerator and denominator expressions are the same.

In general, the limits as $x\to\pm\infty$ of rational expressions can be found by first dividing the numerator and denominator by the highest power of x that appears in the denominator, and then calculating the limit as $x\to\infty$ (or $x\to-\infty$) of both numerator and denominator.

Example 9.23

Calculate $\lim\limits_{x\to\infty}\dfrac{x^3+2x+3}{(5x^2+1)}$.

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